

🔥 Module A: Loss, risk & VC dimension

Big picture Loss measures prediction error; VC dimension measures a classifier class's ability to realize labelings.

Loss functions. For response y and prediction a : squared loss $(y - a)^2$; absolute loss $|y - a|$.

Population risk. $R(f) = \mathbb{E}[L(f(X), Y)]$ averages loss over the population.

Empirical risk. $\hat{R}(f) = n^{-1} \sum_i L(f(x_i), y_i)$ averages loss over training observations. Under squared loss, optimal constants are $\mathbb{E}[Y]$ for population risk and $\bar{y}$ for empirical risk.

Shattering. A class shatters ℓ distinct points if it realizes all 2^ℓ binary labelings of those points.

VC dimension. The largest size of a set the class can shatter; it is infinite if there is no finite bound. To prove $\text{VCdim} = d$, find one shattered d -point set and rule out every $(d + 1)$ -point set. Affine classifiers on $\mathbb{R}^2$ have VC dimension 3.

Training versus test. High training error relative to a suitable benchmark suggests underfitting; low training error with much larger test error suggests overfitting. Test error on an independent sample from the same population estimates population risk. **Recent progress:** Benign overfitting.

⚠️ Remarks

To prove that affine classifiers on $\mathbb{R}^2$ have VC dimension at most 3, show that no set of four points can be shattered. Checking one four-point set is not enough.

🔥 Practice: 1. Bishop 1.26 (squared-loss population risk)

Let $(X, T) \in \mathbb{R}^d \times \mathbb{R}^k$ have a joint density, and let $f : \mathbb{R}^d \rightarrow \mathbb{R}^k$ be a prediction. Define $R(f) = \mathbb{E} \|f(X) - T\|^2$ and $m(x) = \mathbb{E}[T \mid X = x]$, with the needed second moments finite. By expanding the squared norm, derive

$$R(f) = \mathbb{E} \|f(X) - m(X)\|^2 + \mathbb{E} \|T - m(X)\|^2.$$

Hence show that $f(x) = m(x)$ minimizes the population risk.

Hint: fix x , insert and subtract $m(x)$, and use $\mathbb{E}[m(x) - T \mid X = x] = 0$ to eliminate the cross term.

🔥 Practice: 2. ESL 7.8 (VC dimension)

Consider the class of binary classifiers

$$\mathcal{H} = \{h_\alpha(x) = \mathbf{1}\{\sin(\alpha x) > 0\} : \alpha \in \mathbb{R}\}.$$

Here $\mathbf{1}\{A\}$ equals 1 if A holds and 0 otherwise. Show that the points $z_j = 10^{-j}$, $j = 1, \dots, \ell$, can be shattered for every positive integer ℓ , and hence that $\mathcal{H}$ has infinite VC dimension.

Hint: try $\alpha/\pi = 1 + \sum_{i=1}^\ell d_i 10^i$, with $d_i \in \{0, 1\}$. Separate $\alpha z_j/\pi$ into its integer and fractional parts. For an integer k and $0 < r < 1$, $\sin(\pi(k + r))$ has sign $(-1)^k$.

Module B: Least squares & fitted-value geometry

Big picture Minimizing squared error gives normal equations; fitted values are an orthogonal projection.

Design matrix. One row per observation, one column per coefficient. An intercept is a column of ones. The inverse formula requires full column rank.

Dimension. For uniform inputs in a unit cube in $\mathbb{R}^d$, a subcube of side s contains population fraction s^d . For fixed $0 < s < 1$, this decreases as d increases.

Calculus. For symmetric A , $\nabla_w(w^\top Aw) = 2Aw$ and $\nabla_w(a^\top w) = a$. Positive scaling of an objective does not change its minimizers.

Projection. Verify that the fitted vector lies in the column space and its residual is orthogonal to that space.

Practice: 3. Bishop 1.1 (normal equations)

For scalar observations $(x_n, t_n) \in \mathbb{R}^2$, $n = 1, \dots, N$, define

$$g(x, w) = \sum_{j=0}^M w_j x^j, \quad E_D(w) = \frac{1}{2} \sum_{n=1}^N (g(x_n, w) - t_n)^2.$$

Show that the minimizing coefficients solve $\sum_{j=0}^M A_{ij} w_j = T_i$ for $i = 0, \dots, M$, where $A_{ij} = \sum_{n=1}^N x_n^{i+j}$ and $T_i = \sum_{n=1}^N x_n^i t_n$.

Practice: 4. Bishop 3.2 (orthogonal projection)

Let $\Phi \in \mathbb{R}^{N \times p}$ have full column rank, let S be its column space, and let $t \in \mathbb{R}^N$ be the response vector. Show that $H = \Phi(\Phi^\top \Phi)^{-1} \Phi^\top$ projects any vector $v \in \mathbb{R}^N$ orthogonally onto S . Use this to show that the least-squares fitted vector $\Phi \hat{w}$, where $\hat{w} = (\Phi^\top \Phi)^{-1} \Phi^\top t$, is the orthogonal projection of t onto S .